Project Diary for the Software Development Final Year Project (FYP)

BSc (Honours) In Software Development Year Four

A thesisProject presented for the degree of
BSc in Software Development



Computing Department
SETU
Carlow, Ireland
2024

Date:	30/9/2026	Time:	12:15 pm

Weekly Progress report

First meeting: Discussed the scope of the project → A Guitar Chord Detection Tool.

Paul recommended researching similar applications on the market.

FINDINGS: Downloaded many apps.
CHORDIFY has an excellent UI.
ALL APPS have a built-in Guitar Tuner.

COSINE SIMILARITY: Common algorithm referenced by similar applications.
↳ Used to compare a chroma vector against a template vector representing a chord.

Supervisor Comments

Student Signature	Supervisor Signature
M. Lenihan	

Date:	7/10/2026	Time:	12:45

Weekly Progress report

Began discussing First Iteration:

↳ During my Internship I began working on a prototype for the application.

↳ Initially developed in PyCharm, I converted it to a Jupyter Notebook.

↳ A basic Prototype that performs chord analysis using cosine similarity on a pre-recorded clip.

WHAT NEXT → Paul recommend using the Jupyter Notebook for the first demo.

↳ Job for the next week is to figure out recording a clip in-app & THEN perform analysis.

Supervisor Comments

Student Signature	Supervisor Signature
M. Lam	

Date:	16/10/2026	Time:	1:00 pm

Weekly Progress report

Updated via email my progress.

↳ Got a record function working. & demoed it on my laptop mic & a condensor mic.

I also got my wife to demo the chord detection as previously Paul mentioned having an actual beginner interface w/ the algorithm is a truer test.

Suggested making a basic webpage for the first iteration

Paul recommends focusing on the core requirements first & keep it focused in the Jupyter Notebook for the first iteration.

Supervisor Comments

Student Signature	Supervisor Signature
M. Lamb.	

Date:	21/10/2026	Time:	12:15

Weekly Progress report

Made good progress on onset detection. Ran through librosa's tutorial & then applied what I learned.

Currently onset-detect is very sensitive & detects subsequent strums & noise.
I need to use their frame to time function & define a buffer, where anything that happens within the buffer is disregarded.

Paul downloaded my code & ran it over weekend. Had issue w/ channel = 2 / sr2.
Defined a cool target for first release → Display images of three chords & analyse their performance.

Went through prof deliverables
↳ Screencast last (introduce then run code).
↳ Git ✓
↳ Document w/ Readme.
↳ Report

Supervisor Comments

Student Signature	Supervisor Signature